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The Interplay of Niacin Skin Flush Response with Plasma MDA, TNOS, iNOS and Clinical Symptom Severity in Chronic Schizophrenia

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Abstract

Background: Multiple studies have consistently reported elevated oxidative stress and an impaired niacin skin flush response in patients with schizophrenia. This study aimed to compare differences in oxidative stress levels and the niacin skin flush response between patients stratified by the intensity of their response and healthy controls, and to analyze the correlations among oxidative stress markers, the niacin response, and clinical symptoms within the different response subgroups.

Methods: A cross-sectional case-control design was employed. Fifty patients with chronic schizophrenia and 50 demographically matched healthy controls were enrolled. Plasma levels of malondialdehyde (MDA), total nitric oxide synthase (TNOS), and inducible nitric oxide synthase (iNOS) were measured in all participants, and the niacin skin flush response was assessed. Symptom severity in patients was rated using the Positive and Negative Syndrome Scale (PANSS). Based on the intensity of the niacin response, patients were further divided into high-response and low-response subgroups. Regression analysis was used to explore the associations among oxidative stress indicators, the niacin response, and clinical symptoms.

Results: MDA levels were significantly higher in the patient group than in the control group, whereas both TNOS and iNOS levels were significantly lower. The pattern of associations among oxidative stress markers, the niacin skin flush response, and clinical symptoms differed across response subgroups. These relationships may be influenced by factors such as sex, body mass index (BMI), age, and the dosage of antipsychotic medication.

Conclusion: Patients with chronic schizophrenia exhibit abnormalities in oxidative stress and show individual variability in the niacin skin flush response. The intensity of the niacin response may be associated with factors such as sex, BMI, age, and medication, leading to significant differences in the relationships among variables across subgroups in regression analysis. The results suggest that oxidative stress may be involved in the pathophysiological processes of schizophrenia, and that niacin sensitivity may reflect the level of oxidative stress in the body and correlate with different clinical symptomatic profiles.

Keywords

Schizophrenia □ Oxidative stress □ Malondialdehyde □ Nitric oxide synthase □ Niacin skin flush response □ Inflammation

Introduction

Schizophrenia is one of the most common psychiatric disorders, with a global prevalence of approximately 1%. It typically manifests in late adolescence or early adulthood, imposing substantial economic and psychological burdens on patients' families and society. The disorder predominantly follows a chronic course with a high relapse rate.

The diagnosis of schizophrenia is currently based on the ICD-10 or DSM-5 criteria, with core symptoms encompassing positive symptoms (e.g., hallucinations, delusions), negative symptoms (e.g., blunted affect, avolition), and cognitive impairment. Its exact etiology remains incompletely understood, and it is widely regarded as a neurodevelopmental disorder associated with genetic, psychological, and environmental factors. The prevailing hypothesis involves dysfunction in neurotransmitter systems such as dopamine and serotonin, upon which current pharmacological treatments are primarily developed. However, a proportion of patients show poor or even no response to antipsychotic medication, indicating treatment resistance. It is estimated that only about 30% of patients achieve complete symptomatic remission after treatment, primarily in positive symptoms, while negative

symptoms and cognitive impairment are considered the most significant predictors of psychosocial functioning [1, 2] .

Recent research suggests that patients with schizophrenia exhibit an overactive oxidative stress response, indicating its potential involvement in the pathogenesis of the disorder [3] . Imre and Kirkas [4] noted in their study that previous evidence has already indicated the presence of immunonutritional impairment in bipolar disorder. Combined with their own findings in schizophrenia, inflammation and oxidative stress may play shared pathophysiological roles across psychiatric disorders. Therefore, investigating such markers not only contributes to understanding the pathological mechanisms of schizophrenia but also offers new perspectives on the common mechanisms underlying multiple mental disorders. Although no validated biomarkers are currently available for clinical stratification or treatment outcome prediction, related investigations are actively ongoing.

Measuring the plasma levels of malondialdehyde (MDA), total nitric oxide synthase (TNOS), and inducible nitric oxide synthase (iNOS) in patients aids in assessing the oxidant-antioxidant imbalance and inflammatory status. As a primary end product of lipid peroxidation, the concentration of MDA directly reflects the level of oxidative stress. Studies have confirmed that MDA levels are significantly

abnormal in patients with schizophrenia across different illness stages and subtypes [5-8]. Nitric oxide synthase (NOS) catalyzes the production of nitric oxide (NO). Among its isoforms, the inducible form (iNOS) is upregulated under pathological conditions such as acute stress or chronic inflammation, serving as a biomarker for inflammation and tissue damage. Research indicates that NOS levels are abnormal in both the peripheral blood and brain tissues of patients with schizophrenia [9-13].

The niacin skin flush response is triggered by the activation of the cutaneous GPR-109A receptor by niacin, leading to the production of prostaglandin E₂ (PGE₂) via the phospholipase A₂ → PLA₂ → arachidonic acid → AA → cyclooxygenase-1 (COX-1) pathway, which induces vasodilation [14]. An attenuated response has been repeatedly demonstrated in patients with schizophrenia [15-18], possibly indicating abnormalities in the signaling pathway, functional deficits of key enzymes, or reduced local vascular reactivity [19-21]. Methods for assessing this response include the oral niacin challenge [22], visual rating scales [23], and laser Doppler flowmetry [24]. The present study employed a computer-based quantitative scoring method to minimize subjective bias. Reduced niacin sensitivity is associated with greater functional impairment, abnormal protein expression, and inflammatory

dysregulation in patients [25, 26]. Furthermore, arachidonic acid levels in erythrocyte membranes are generally lower in patients than in healthy controls [27, 28]. Consequently, an attenuated niacin response is considered a potential endophenotype that may aid in differential diagnosis [24, 29].

Based on this background, the present study employed a case-control design. It aimed to compare the differences in oxidative stress markers and the niacin skin flush response between chronic schizophrenia patients with distinct niacin response profiles and healthy controls, and to further explore the correlations among MDA, TNOS, iNOS, the niacin response, and clinical symptoms. This work was conducted to provide preliminary evidence for elucidating their underlying relationships.

Subjects and methods

Study Participants

The inclusion criteria for the schizophrenia group (SZ group) were as follows: (a) meeting the diagnostic criteria for schizophrenia according to the *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5); (b) age between 18 and 65 years;

and (c) hospitalization for ≥ 1 year, a total illness duration of ≥ 2 years, and currently being in a clinically stable phase.

The healthy control group (HC group) comprised 50 individuals recruited via community advertisements in Suzhou, who were matched to the patient group in terms of age, sex, and body mass index (BMI). Their inclusion criteria were: (a) no personal history of psychiatric disorders; and (b) age between 18 and 65 years.

Exclusion criteria for both the schizophrenia group and the healthy control group included: (a) comorbid neurological disorders (e.g., epilepsy, brain tumors, or a history of severe brain injury); (b) use of non-steroidal anti-inflammatory drugs or corticosteroids within the 14 days prior to the study; (c) pregnancy or lactation; and (d) a history of alcohol or substance abuse. All participants provided written informed consent after receiving a comprehensive explanation of the study. The study protocol was approved by the Ethics Committee of Suzhou Guangji Hospital (Approval No.: 2019-051).

General information from patients, including sex, age, body mass index (BMI), and medication details, was collected using standardized questionnaires. Antipsychotic dosages were uniformly converted to chlorpromazine equivalent doses for assessment. The

severity of clinical symptoms was evaluated by two trained psychiatrists using the Positive and Negative Syndrome Scale (PANSS). The inter-rater reliability for PANSS scores, expressed as the correlation coefficient (r), exceeded 0.8.

Blood Sample Collection and Biochemical Assays

Blood samples were collected from the antecubital vein from all participants after an overnight fast (10–12 hours). Following anticoagulation treatment, the samples were centrifuged at 3000 r/min for 15 minutes to separate the plasma. The supernatant plasma was aspirated using a pipette, aliquoted into 500 μ L microcentrifuge tubes, and stored at -80°C until batch analysis.

Plasma levels of MDA (catalog no. A003-1), TNOS and iNOS (catalog no. A014-2) were measured with the assistance of the Nanjing Jiancheng Bioengineering Institute. MDA was quantified using the thiobarbituric acid (TBA) method, while TNOS and iNOS were determined by colorimetric assays. Results are expressed as nmol/L for MDA and U/mL for TNOS and iNOS.

Measurement of TNOS and iNOS activities

Total nitric oxide synthase (TNOS) and inducible NOS (iNOS) activities in plasma were determined using a commercial

colorimetric assay kit (Nanjing Jiancheng Bioengineering Institute, China, catalog No. A014-2) according to the manufacturer's instructions. The principle was based on the catalytic conversion of L-arginine and molecular oxygen to nitric oxide (NO) by NOS; NO then reacted with a nucleophilic substance to form a colored compound with a characteristic absorption peak at 530 nm. The absorbance at 530 nm was measured, and the enzyme activity was calculated accordingly.

One unit (U) of enzyme activity was defined as the amount of enzyme that produced 1 nmol of NO per minute per milliliter of serum. TNOS/iNOS activity (U/mL) was calculated using the following formula:

$$\text{Activity (U/mL)} = \frac{OD_{\text{assay}} - OD_{\text{blank}}}{\varepsilon} \times \frac{V_t}{V_s} \times \frac{1}{l \times t} \div 1000$$

In the formula for enzyme activity (U/mL), OD_{assay} and OD_{blank} denote the absorbances of the sample and corresponding blank tubes, ε is the nanomolar extinction coefficient (provided by the kit), V_t and V_s are the total reaction and sample volumes (mL), l is the cuvette path length (1 cm), and t is the reaction time (min).

Division by 1000 converts the value to U/mL.

Niacin Skin Flush Test

In the present study, the instrument used to evaluate the cutaneous

flushing response to niacin was provided by Shanghai Tianyin Biotechnology Co., Ltd. (model number: TY-Lab-01). This device consists of three main components: a six-well standard patch, an automated imaging system, and intelligent image recognition software (patent number: ZL 20211070775.0). Six different concentrations of nicotinic acid methyl ester solution were prepared as threefold serial dilutions, starting from 60 mmol/L. Specifically, the first concentration was 60 mmol/L, followed by 20 mmol/L, 6.67 mmol/L, 2.22 mmol/L, 0.74 mmol/L, and finally 0.25 mmol/L. Equal volumes of each concentration were applied to the corresponding wells of the patch. The application sites on the subject's right forearm were arranged linearly: the most distal site was designated as the first point, with the second, third, fourth, fifth, and sixth points placed progressively more proximal. The patch was placed in direct contact with the subject's right forearm for one minute and then removed. Immediately after removal, the forearm was positioned and secured within the testing apparatus. An automated imaging system captured photographs of the right forearm at ten-second intervals over a ten-minute period, yielding a total of 60 images per subject. The intelligent image recognition software was then used to identify the skin areas that had been in contact with the nicotinic acid methyl ester solution in each

photograph. The image taken at the 10-second time point was algorithmically defined as the baseline reference representing the skin's initial state; consequently, the remaining 59 images were included in the final analysis. For each subject, this process generated 354 erythema area values (59 time points $\times$ 6 concentrations). The software automatically identified and calculated the erythema area at each captured time point and subsequently summed the total area across different time points [30].

For some participants, the flushing response at concentration levels 4, 5, and 6 was so weak that the computer software could not reliably delineate the erythema area at these sites. Consequently, these measurements were recorded as missing values. To preserve the authenticity of the data and the accuracy of the study findings, the total area values corresponding to these three concentration points were omitted from the quantitative analysis, as presented in Table 1.

Statistical Analysis

Data Description and Normality Testing

Continuous variables were first assessed for normality using the

Shapiro–Wilk test. Data conforming to a normal distribution are presented as mean $\pm$ standard deviation (mean $\pm$ SD), while non-normally distributed variables are reported as median (interquartile range) [M (IQR)]. Categorical variables are summarized as frequency (percentage).

Methods for Between-Group Comparisons

Based on the distribution characteristics of the variables and the number of groups, a tiered testing strategy was adopted:

- **Two-group comparisons:** Normally distributed variables were analyzed using the independent samples *t*-test; non-normally distributed variables were compared using the *Mann-Whitney U* test.
- **Overall comparisons across three or more groups:** Normally distributed variables were examined using one-way analysis of variance (one-way ANOVA); non-normally distributed variables were analyzed with the *Kruskal-Wallis H* test.
- **Post hoc pairwise comparisons:** If the overall ANOVA result was significant ($P < 0.05$), *Tukey's HSD* test was applied for post hoc analysis. If the overall *Kruskal-Wallis* test was significant, the *Dunn* test was used for pairwise

comparisons, with P -values adjusted using the Bonferroni correction to control for multiple comparisons.

□ **Categorical variable comparisons:** The χ^2 test was used.

Correlation Analysis and Regression Modeling

Based on the results of normality testing, Pearson correlation analysis was used for normally distributed variables, and Spearman's rank correlation analysis was used for non-normally distributed variables to preliminarily examine bivariate associations.

To further quantify the relationships between variables while controlling for potential confounding factors, regression models were subsequently constructed. When building the regression models, sex, BMI, age, and antipsychotic dosage (converted to chlorpromazine equivalents) were included as covariates regardless of the significance of the bivariate correlations, in order to account for their potential influence. The linearity assumption of the relationships was first assessed. If the relationship was approximately linear and the residuals satisfied the model assumptions, least-squares linear regression was applied.

Effect Sizes and Significance Level

For results reaching statistical significance, corresponding effect sizes are reported. All tests were two-tailed, with the significance

level set at $\alpha = 0.05$. Data analysis was performed using SPSS version 27.0.

Results

Comparison of Demographic and General Clinical Characteristics

Based on the median total niacin flush area at the first concentration point at 10 minutes (denoted as 10min-1, and so forth for other time points) in the patient group, patients were divided into a low-response group and a high-response group (each $N = 25$) for subsequent analysis.

As shown in Table 1, this study compared differences across multiple variables among the three groups (low-response, high-response, and healthy control groups). Overall, the groups were well-matched in baseline characteristics such as sex, age, and body mass index (BMI), with no statistically significant differences. Between the two patient subgroups, no significant differences were observed in the severity of clinical symptoms (scores on various scales) or in antipsychotic dosage. Among the high-response group, 8 patients received atypical antipsychotic monotherapy, and 17 patients received combination therapy with different antipsychotics. In the low-response group, 9 patients received atypical antipsychotic monotherapy, and 16 patients received combination therapy with different antipsychotics. In each group, only one patient used a typical antipsychotic (perphenazine) as part of the

combination regimen. Detailed information is presented in Table 1.

Oxidative stress markers (MDA, TNOS, iNOS) and the total area of the niacin skin flush response at different time points and concentrations all showed significant differences among the three groups. Specifically:

- MDA levels were significantly higher in both patient groups than in the healthy control group.
- TNOS and iNOS levels were highest in the healthy control group, lower in the patient groups, and higher in the low-response group than in the high-response group.
- Regarding the niacin flush area, except for the second concentration point at 10 minutes (10min-2), where results showed healthy controls > low-response group and high-response group > healthy controls, the results for the remaining time-concentration points indicated that both the healthy controls and the high-response group had significantly larger flush areas than the low-response group, with no significant difference between the healthy controls and the high-response group.

The baseline characteristics of the three groups and the results of the between-group comparisons for the main study variables are detailed in Table 1. Figure 1 presents a comparison of TNOS and

iNOS activities among the three groups.

Table 1. Demographic characteristics and general clinical data across the three groups.

	Healthy Controls (<i>n</i> =50) Group A	Low-Response Group (<i>n</i> =25) Group B	High-Response Group (<i>n</i> =25) Group C	Test Statistic	<i>P</i> (Overall)	<i>P</i> (Post Hoc)
Demographics						
Age [years]	45 (39.75, 50.25)	50.68 ± 9.945	48.76 ± 9.185	$H(2)=5.417$	0.067	—
Sex [n, %]				$\chi^2=3.526$	0.172	—
Male	22 (44)	16 (64)	10 (40)			
Female	28 (56)	9 (36)	15 (60)			
Education [years]	15 (12, 16)	12 (9, 12)	12 (9, 12)	$H(2)=29.937$	<.001**	A-B: <.001**; A-C: <.001**; B-C: 1.000
BMI [kg/m ²]	21.240 ± 1.438	21.481 ± 1.161	21.586 ± 1.333	$F(2)=0.632$	0.534	—
Duration of illness (year)	—	15.28±7.334	15.44±6.758	$t=-0.08$	0.936	—
Oxidative Stress Markers						
MDA (nmol/L)	2.389 (1.536, 3.191)	4.096 ± 0.945	3.077 (2.922, 3.831)	$H(2)=33.259$	<.001**	A-B: <.001**; A-C: <.001**; B-C: 0.307
TNOS [U/mL]	11.998 ± 1.391	10.078 ± 1.341	8.673 ± 0.961	$F(2)=59.575$	<.001**	A-B: <.001**; A-C: <.001**; B-C: <.001**
iNOS (U/mL)	8.133 ± 0.903	7.162 ± 1.099	5.971 ± 0.787	$F(2)=45.757$	<.001**	A-B: <.001**; A-C: <.001**; B-C: <.001**
Niacin Flushing Area						
5min-1	1664.486 ± 523.889	870.184 ± 402.639	1736.936 ± 324.386	$F(2)=30.879$	<.001**	A-B: <.001**; A-C: 1.000; B-C:

						<.001**
5min-2	1272.902 ± 526.644	489.902 ± 384.629	1390.300 ± 366.588	$F(2)=30.72$ 7	<.001**	A-B: <.001**; A- C: 0.895; B-C: <.001**
5min-3	952.276 ± 549.439	227.3 (0, 443.45)	898.440 ± 432.769	$H(2)=29.75$ 8	<.001**	A-B: <.001**; A- C: 1.000; B-C: <.001**
10min-1	4046.7 (3604.63, 4838.98)	2891.3 (2025.6, 3210.15)	4266.516 ± 546.747	$H(2)=42.97$ 5	<.001**	A-B: <.001**; A- C: 1.000; B-C: <.001**
10min-2	3256.688 ± 1093.629	1719.102 ± 930.559	3666.692 ± 582.028	$F(2)=30.67$ 9	<.001**	A-B: <.001**; A- C: <.001**; B-C: 0.109
10min-3	2789.386 ± 1130.248	1199.884 ± 867.445	2831.452 ± 722.088	$F(2)=25.11$ 3	<.001**	A-B: <.001**; A- C: 1.000; B-C: <.001**
PANSS Scores	□	□	□	□	□	□
Positive (P)	—	14 (10, 16.5)	14 (10.5, 23)	$U=273.5$, $r=-0.107$	0.455	—
Negative (N)	—	19.84 ± 7.122	15 (13.5, 22)	$U=251.5$, $r=-0.168$	0.240	—
General (G)	—	33.16 ± 7.872	34 (28.5, 43.5)	$U=262.5$, $r=-0.137$	0.337	—
Total	—	64 (54, 78.5)	70 (53.5, 89)	$U=262$, $r=-0.139$	0.332	—
Medication	□	□	□	□	□	□
CPZeq□mg/day□	—	433.366 ± 135.968	393.160 ± 182.048	$t(48)=0.26$ 5	0.792	—
Atypical antipsychotics alone (n,%)						
Aripiprazole	—	2(8)	0			
Olanzapine	—	3(12)	2(8)			
Clozapine	—	3(12)	4(16)			
Risperidone	—	1(4)	1(4)			
Quetiapine	—	0	1(4)			

A combination of**antipsychotics****(n,%)**

Olanzapine- risperidone	—	4(16)	3(12)
Quetiapine- perphenazine	—	1(4)	1(4)
Risperidone- quetiapine	—	1(4)	5(20)
Olanzapine- aripiprazole	—	2(8)	2(8)
Risperidone- aripiprazole	—	2(8)	3(12)
Clozapine- risperidone	—	4(16)	1(4)
Clozapine- aripiprazole	—	2(8)	2(8)

BMI, body mass index;

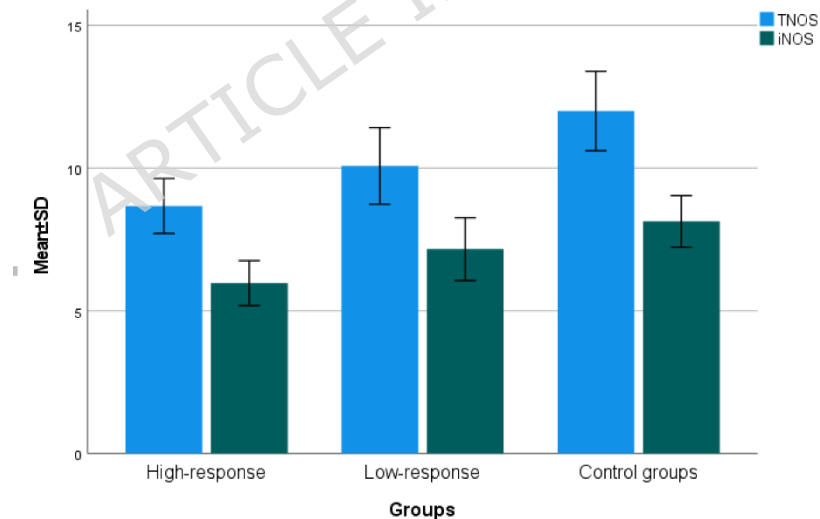
PANSS, Positive and Negative Syndrome Scale;

CPZeq, chlorpromazine equivalent dose;

* $P < 0.05$; ** $P < 0.001$;

"—" indicates not applicable;

5min-1, total niacin-induced flushing area at the first concentration point measured at 5 min.

**Figure 1.** Mean $\pm$ SD of TNOS and iNOS in three groups (clustered bars).

Correlation Analysis: Relationships among Oxidative Stress Markers, Clinical Symptoms, and Niacin Skin Flush Response Area in Different Patient Subgroups

The analysis revealed the following correlations:

□ In the **low-response subgroup**:

- The flush area at 10min-1 was negatively correlated with positive symptom subscale scores ($r = -0.42$, $P = 0.037$).
- TNOS levels were negatively correlated with both PANSS total scores ($r = -0.447$, $P = 0.025$) and general psychopathology subscale scores ($r = -0.488$, $P = 0.013$).
- iNOS levels were also significantly negatively correlated with general psychopathology subscale scores ($r = -0.409$, $P = 0.042$).

□ In the **high-response subgroup**:

- The flush area at 10min-1 was significantly positively correlated with negative symptom subscale scores ($r = 0.611$, $P = 0.001$).
- The flush area at 5min-1 was significantly positively correlated with iNOS levels ($r = 0.398$, $P = 0.049$).
- The flush area at 5min-3 showed a significant positive correlation with general psychopathology subscale scores ($r = 0.408$, $P = 0.043$).

Detailed results are presented in Table 2.

Table 2. Associations among oxidative markers, symptoms, and niacin response by patient subgroup.

Group	Variable 1	Variable 2	<i>R</i>	<i>P</i>
Low-response	10min-1	P subscores	-0.420	0.037*
	TNOS	PANSS total scores	-0.447	0.025*
	TNOS	G subscores	-0.488	0.013*
	iNOS	G subscores	-0.409	0.042*
High-response	10min-1	N subscores	0.611	0.001**
	5min-1	iNOS	0.398	0.049*
	5min-3	G subscores	0.408	0.043*

Regression Analysis

Building on the aforementioned correlation analyses, we performed multiple linear regression to construct the core analytical models. Although sex, age, BMI, and chlorpromazine-equivalent antipsychotic dosage showed no significant differences between the two patient subgroups, these variables were nevertheless included as covariates in the final regression models for adjustment, given that previous literature suggests their potential influence on the outcomes [18, 31-34]. This approach aimed to obtain a more precise estimate of the independent effects of the target variables and to reduce potential omitted variable bias.

The results indicated that statistically significant findings were exclusively observed in the high-response subgroup. In the low-response subgroup, all correlations that were significant in Table 2 became attenuated and lost statistical significance after adjusting for the aforementioned covariates in the regression models.

Notably, within the high-response subgroup, several variable pairs that showed no significant correlation in the bivariate analysis demonstrated statistically significant changes in their regression relationships after controlling for these covariates. Detailed results are presented in Table 3.

Table 3. Multiple linear regression models.

Group	Model Specification (<i>DV</i> ~ <i>IV</i>)	β	<i>t</i>	<i>P</i>	R^2 / <i>Adj. R</i> ²
Low-response	TNOS ~ PANSS total scores	-0.169	-0.863	0.399	0.412 / 0.257
	TNOS ~ G subscores	-0.326	-1.647	0.116	0.401 / 0.244
	iNOS ~ G subscores	-0.264	-1.328	0.200	0.374 / 0.209
	10min-1 ~ P subscores	0.060	0.288	0.777	0.439 / 0.291
High-					

response

iNOS ~ 5min-1	0.352	1.709	0.104	0.234 / 0.033
5min-3 ~ G subscores	-	-	0.232	0.432 / 0.221 1.235 0.283
TNOS ~ 5min-3	0.481	2.404	0.027*	0.281 / 0.092
N subscores ~ 10min-1	0.468	2.227	0.038*	0.268 / 0.075
TNOS ~ PANSS total scores	-	-	0.032*	0.523 / 0.377 2.314 0.397
TNOS ~ G subscores	-	-	0.041*	0.511 / 0.363 2.197 0.382
iNOS ~ G subscores	-	-	0.018*	0.548 / 0.412 2.601 0.429
5min-3 ~ iNOS	0.469	2.295	0.033*	0.256 / 0.061
iNOS ~ PANSS total scores	-	-	0.006*	0.595 / 0.466 3.112 0.488

Discussion

This study demonstrates that patients with chronic schizophrenia exhibit two distinct pathological manifestations: elevated levels of the oxidative stress product MDA, and decreased levels of TNOS and iNOS. Furthermore, significant differences were observed in the area of the niacin skin flush response among different patient

subgroups. These findings show some inconsistencies with previous studies concerning oxidative stress and the niacin skin response [33, 35, 36].

The core finding of this study is that, among chronic schizophrenia patients with different types of niacin response, there exist correlations of varying strength and direction between nitric oxide synthase (TNOS/iNOS), the area of the niacin skin flush response, and the severity of clinical symptoms. Additionally, this study did not find a direct association between the level of the oxidative stress marker MDA and either clinical symptoms or the niacin skin response. This differs from some prior research, suggesting that in this chronic patient population, MDA levels may represent a relatively independent pathological dimension.

The present study found that peripheral plasma MDA levels were significantly higher in patients with chronic schizophrenia than in healthy controls. This result is consistent with the conclusions of the majority of previous studies, which have reported elevated MDA in first-episode, drug-naïve patients [5], chronically medicated patients [33, 37], and those with treatment resistance [6]. However, some studies focusing on first-episode, drug-naïve and chronic patients have reported contradictory findings [38, 39]. Research by

Yang et al. [40] suggests that MDA levels in patients may decrease following antipsychotic treatment.

The changes in TNOS and iNOS levels observed in this study are consistent with some existing reports, such as those by Malik and Yang et al. [13, 41], but inconsistencies also exist. For example, Taneli et al. [42] found that serum NO levels in schizophrenia patients remained higher than those in healthy controls and showed no significant change even after 6 weeks of antipsychotic treatment. Furthermore, studies on postmortem brain tissue from patients indicated significantly higher neuronal NOS expression in the frontal cortex compared to controls [43]. Research suggests that NO at low to moderate concentrations may exert neuroprotective effects, whereas excessively high concentrations can be neurotoxic [10]. Excessive NO can induce oxidative stress, impair neuronal function, and lead to cell death, indicating that overproduction of NO may be related to the pathological mechanisms of schizophrenia. Animal model studies provide further support for this view: nNOS knockout mice exhibit upregulated dopamine D1 receptor signaling, accompanied by abnormal social behavior, hyperactivity, and impaired spatial memory [44]; concurrently, NOS inhibitors can block the behavioral and

biochemical alterations induced by PCP in rats [45].

The inconsistencies among the aforementioned study results may be related to differences in patients' medication status, illness duration, and other factors. Previous research indicates that antipsychotic medications themselves can influence inflammatory and oxidative stress systems. Different types and doses of drugs may have bidirectional regulatory effects on iNOS expression, with some potentially inducing inflammation in the short term and others suppressing it. For example, Nonaka-Hashida et al. [46] found that antipsychotics could reduce elevated methylated arginine levels and increase L-arginine in patients with schizophrenia, thereby improving NO bioavailability. Animal experiments also suggest that cholinergic receptor antibodies derived from patients can interact with brain M1 receptors, triggering the production of NO, PGE₂, and MMP-3, and inducing the expression of COX-1 and iNOS mRNA in the prefrontal cortex—effects that can be inhibited by haloperidol [47]. Furthermore, olanzapine has been shown to attenuate astrocyte proliferation in the rat prefrontal cortex and downregulate the expression of related pro-inflammatory, oxidative stress, and apoptotic molecules [48]; risperidone can reduce iNOS expression, suggesting it

possesses certain anti-inflammatory potential [49] .

In summary, the inconsistencies in MDA and NOS findings may reflect dynamic changes in oxidative stress and inflammation levels in patients with schizophrenia, such as elevation during acute phases and a gradual decrease in chronic stages. In this study of chronic schizophrenia patients, correlation analyses revealed no significant associations between MDA levels and clinical symptoms or the niacin skin flush area. The lack of significant findings may be related to the relatively small sample size, although this possibility cannot be ruled out. Considering the high degree of heterogeneity inherent to schizophrenia itself, the results of this study also indirectly highlight the complexity of the disorder and the potential methodological differences across studies. Chronic patients predominantly present with negative symptoms, and their neuroimmunopathological mechanisms may differ from those in the acute phase or in patients with shorter illness durations. This aspect warrants further in-depth investigation in the future.

The present study indicates significant associations between TNOS and iNOS levels and different dimensions of clinical symptoms. Yang et al. [13] reported a positive correlation between plasma NO

levels and PANSS positive symptom scores; McNeill [50] proposed that elevated peripheral NO metabolite concentrations may be associated with enhanced inflammatory activity in psychopathology. Animal experiments further suggest a link between dopamine dysfunction-related cognitive deficits and NO [51], and that NOS inhibitors can attenuate schizophrenia-like effects induced by NMDA receptor hypofunction. When NOS enzyme activity becomes excessively elevated, oxidative stress processes intensify, potentially leading to aggravated cellular damage; conversely, effective management of psychiatric symptoms may help mitigate such stress responses. We hypothesize that the observed decrease in NOS levels in patients with chronic schizophrenia may reflect a degree of regulation in their neuroinflammatory state. Imre et al. [52] reported elevated inflammatory markers in schizophrenia but found no association with cognition. In contrast, our MDA and NOS indices reflected oxidative stress and abnormal inflammation-endothelium interactions. Therefore, a multidimensional biomarker approach integrating inflammation, oxidative damage, and vascular function offers greater value for elucidating pathophysiological mechanisms and for risk stratification.

Although the different niacin skin flush response groups showed no

significant differences across clinical symptom dimensions, this study still identified correlations between response intensity and clinical symptoms. Notably, after controlling for covariates, regression analysis revealed a differential pattern. In the low-response group, most bivariate correlations became non-significant after multivariate adjustment. In the high-response group, however, some initially uncorrelated variables emerged as significant predictors following adjustment. This important finding suggests that the relationships among variables in different response groups may be mediated or confounded by the control variables in the model (sex, age, BMI, and chlorpromazine-equivalent antipsychotic dosage). It also indicates that differences in niacin sensitivity may partially reflect variations in the severity of clinical symptoms and the level of systemic inflammation.

Metabolomic and proteomic studies suggest that patients with schizophrenia exhibit impairments in energy metabolism and lipid biosynthesis [53]. Abnormalities in phospholipid metabolism, in particular, can be manifested and detected through an attenuated niacin skin flush response. As summarized by Messamore, the abnormal niacin response in schizophrenia is independent of psychotropic medication use and may indicate a deficiency in

essential fatty acids [54]. Research by Chen et al. further notes that the attenuation of the niacin skin flush response is not associated with illness severity, violent behavior, or antipsychotic dosage, suggesting its potential as a trait marker for schizophrenia and supporting its utility as an objective diagnostic biomarker for distinguishing patients from healthy controls [55]. A study by Tavares et al. demonstrated elevated phospholipase A2 (PLA2) activity in schizophrenia patients, with higher activity observed in niacin non-responders than in responders; after 8 weeks of antipsychotic treatment, this activity decreased [31]. This may help explain why a subset of patients exhibits a niacin response similar to that of healthy controls.

Correlational studies have also shown that in individuals at ultra-high risk and in first-episode psychosis patients, niacin sensitivity is inversely correlated with positive symptom scores [56]. Zhang et al. reported that an attenuated niacin response in first-episode psychosis patients is significantly associated with negative symptoms, indicating that the absence of a niacin response could serve as a potential biomarker for a subtype characterized predominantly by negative symptoms and poor treatment response [57]. Notably, it has recently been reported that several cardiovascular risk markers exhibit significant differences among

different clinical subgroups of patients with schizophrenia, suggesting that biomarker-based stratification strategies are emerging as a new direction in psychiatric research [58]. In conjunction with our results, niacin response—an endophenotypic indicator reflecting oxidative stress-inflammation pathways—may in the future be combined with other biomarkers (e.g., metabolic and inflammatory markers) to facilitate patient risk stratification or the selection of individualized treatment regimens.

Currently, the variety of methods used to assess the niacin skin flush response hampers direct comparability across studies, and differences in outcome criteria may lead to variations in reported significance. There is a need to establish more standardized measurement protocols in the future to overcome the current limitations in comparability. The findings of this study reinforce the view that multiple variables often interact with sex, age, BMI, and antipsychotic medication. This highlights the importance of comprehensively considering these factors in future research and offers insights for developing more targeted intervention strategies.

Limitations

This study has several limitations. First, as a cross-sectional study, its design cannot capture the dynamic evolution of the relevant

indicators, and the conclusions reflect characteristics only at a specific time point. Future research should include patients at different illness stages and with varied clinical profiles to conduct longitudinal follow-up for more in-depth exploration. Second, this study did not perform subgroup analyses or statistical control based on the specific types of antipsychotic medications. Given that different medications may exert differential effects on the NOS system, oxidative stress, and the niacin skin flush response, future studies should refine their design to control for additional potential confounding factors. These refinements may include, but are not limited to, stratifying by the type of antipsychotic used, incorporating a longitudinal time axis, improving within-subject comparisons, and dynamically tracking and exploring the correlations among the various indicators. Third, the sample sizes of some subgroups in this study were relatively small (e.g., $n=25$), which may lead to insufficient power in the regression analyses, causing some true associations to fail to reach statistical significance, and may also render the observed significant associations unstable. Therefore, these regression results should be regarded as exploratory and preliminary, and need to be validated in larger samples in future research. Fourth, this study does not offer a novel pathophysiological or therapeutic insight into

oxidative stress or the niacin flush response in schizophrenia. Limitations include the lack of large-scale Chinese data, the inclusion of only stable-phase patients, and unstandardized assessment methods. Despite computer-based quantification to enhance objectivity, this study remains a confirmatory, rather than a conceptual, contribution to existing evidence in Chinese stable-phase patients.

Conclusion

In summary, this preliminary study suggests that TNOS and iNOS levels in patients with schizophrenia may be potentially associated with the severity of their clinical symptoms and the sensitivity of the niacin skin response. These findings further support the notion that NOS may be involved in the abnormal process of the niacin skin flush response in schizophrenia, while also indicating that oxidative stress and inflammatory responses may be related to the pathophysiological mechanisms of the disorder. The sensitivity of the niacin response may reflect the body's oxidative stress status and show connections with different clinical symptomatic profiles. From a clinical translation perspective, the aforementioned biomarkers could be used for patient risk stratification (e.g., identifying subgroups with higher oxidative stress levels) and provide a basis for selecting targets for adjunctive therapies directed at oxidative stress, such as antioxidants and anti-inflammatory agents. We hope this study can provide partial theoretical support for the development of novel adjuvant

therapeutic strategies targeting oxidative stress and inflammation, and offer insights for improving the clinical prognosis of schizophrenia.

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Author contributions

Lihua Chen wrote the manuscript; Xiaobin Zhang was responsible for study design; Jiyong He performed the statistical analysis; Lihua Chen and Jiyong He were responsible for recruiting the patients and collecting the samples. All authors have contributed to and have approved the final manuscript.

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Data availability

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval and consent to participate

We declare that all experiments on human subjects were conducted in accordance with the Declaration of Helsinki and that all procedures were carried out with the adequate understanding and written consent of the subjects. All experimental protocols were approved by the Ethics Committee of The Affiliated Guangji Hospital of Soochow University. Informed consent was obtained from all the participants and/or their legal guardians. All methods were carried out in accordance with relevant guidelines and regulations.

Consent for publication

Not applicable.

Conflict of interest

The authors declare no conflict of interest.

Competing interests

The authors declare no competing interests.

Clinical trial

No applicable.

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